

(Non-asymptotic)
Spectral Properties of Heavy-Tailed Random
Matrices

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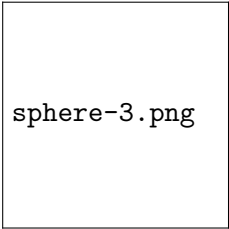
March 26, 2019

Non-asymptotic...

$A = (A_{ij})_{n \times m}$. A_{ij} are taken from some distribution.

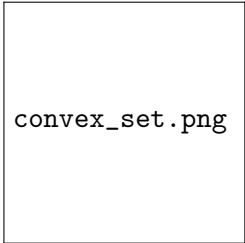
Usually, we have

- no specific distribution assumption
- no symmetry assumption
- high probability results (hold with probability $1 - o(1)$)
- for the dimensions large enough (all large matrices with $n, m > N_0$)



sphere-3.png

Concentration on the sphere



convex_set.png

Convex set in $\mathbb{R}^n$

Spectral properties...

Singular values – real spectrum of the matrix

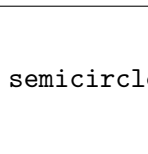
$$s(A) = \sqrt{\text{eig}(A^T A)}, \quad s_1 \geq s_2 \geq \dots \geq s_n \geq 0.$$

Why consider singular values?

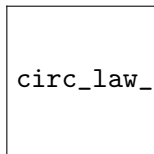
operator (spectral) norm $\|A\| = \sup_{\|x\|_2=1} \|Ax\|_2 = s_1(A)$

norm of the inverse $1/\|A^{-1}\| = \inf_{\|x\|_2=1} \|Ax\|_2 = s_n(A)$

Key idea: spectrum stabilizes as the size of the matrices $\rightarrow \infty$



semicircle-unscaled.jpg



circ_low_3.png

(Left picture is taken from "Estimation in high dimensions by R.Vershynin")

Light tails

Let $A = (A_{ij})_{n \times n}$ be a square random matrix with i.i.d. entries.

Definition (subgaussian entries)

A_{ij} are subgaussian if $\mathbb{P}\{|A_{ij}| > t\} \leq C_1 e^{-c_2 t^2}$ for any $t > 0$

	Gaussian	Subgaussian
$s_1(A)$	$\mathbb{P}\{s_1 > 2\sqrt{n} + t\} < 2e^{-t^2/2}$ $t \geq 0$ from Gordon's theorem	$\mathbb{P}\{s_1 > t\sqrt{n}\} \leq e^{-ct^2n}$ $t \geq C_0$ from Bernstein's inequality
$s_n(A)$	$\mathbb{P}\{s_n \leq \frac{\varepsilon}{\sqrt{n}}\} < \sqrt{2}e^{-\varepsilon}$ $\varepsilon \geq 0$ Edelman, Szarek	$\mathbb{P}\{s_n \leq \frac{\varepsilon}{\sqrt{n}}\} \leq C_1\varepsilon + e^{-c_2n}$ A_{ij} mean zero, $\varepsilon \geq 0$ Rudelson-Vershynin

So, for light-tailed random matrices: with high probability,

$$\|A^{-1}\| = 1/s_{\min}(A) \sim \sqrt{n} \quad \text{and} \quad \|A\| = s_{\max}(A) \sim \sqrt{n}.$$

Heavy tails

Light tails: with high probability,

$$\|A^{-1}\| = 1/s_{\min}(A) \sim \sqrt{n} \quad \text{and} \quad \|A\| = s_{\max}(A) \sim \sqrt{n}.$$

Heavy tails (two finite moments): with high probability,

$$\|A^{-1}\| = 1/s_{\min}(A) \sim \sqrt{n} \quad \text{and} \quad \|A\| = s_{\max}(A) \not\sim \sqrt{n}.$$

Example ($\|A\| \sim n \gg \sqrt{n}$)

Litvak-Spector: Constructive example of $\|A\| \sim O(n^{1-\beta})$ for any $\beta \geq 0$ with probability at least $1/2$.

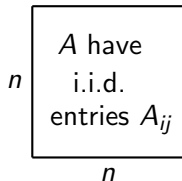
Can be also concluded from Bai-Silverstein-Yin results: 4 moments are needed for $\|A\| \sim \sqrt{n}$.

NEXT: OVERVIEW OF THE RESULTS

1. Invertibility: lower bound for $s_{\min}(A)$
2. Norm regularization: upper bound for $s_{\max}(A)$
3. Constructive regularization:
 - a) case 1: more than two finite moments
 - b) case 2: Bernoulli matrices (and random graphs)
 - c) case 3: general case

1. Invertibility

Random matrix A



Rudelson-Vershynin:

if A_{ij} **mean zero subgaussian**,

$$\mathbb{P}\{s_{\min}(A) \leq \varepsilon n^{-1/2}\} \leq L\varepsilon + e^{-un}$$

$L, u > 0$ are constants that depend only on subgaussian constants in the distribution of A_{ij}

Theorem (Invertibility; with K. Tikhomirov)

Let $A = (A_{ij})_{n \times n}$ has i.i.d. entries, s.t. $\mathbb{E}A_{ij} = 0$, $\mathbb{E}A_{ij}^2 = 1$:

$$\mathbb{P}\{s_{\min}(A) \leq \varepsilon n^{-1/2}\} \leq L_1\varepsilon + e^{-u_1 n} \text{ for any } \varepsilon \in (0, 1)$$

$L_1, u_1 > 0$ are constants depending only on the distribution of A_{ij} .

Random ellipsoid coverings

Theorem (Universal covering for ellipsoids, informal statement)

Let $A = (A_{ij})_{n \times n}$ has i.i.d. entries, s.t. $\mathbb{E}A_{ij} = 0$, $\mathbb{E}A_{ij}^2 = 1$.

$$B_2^n := \text{unit ball in } \mathbb{R}^n;$$

$$\text{random ellipsoid } \mathcal{A} := \{Ax : x \in B_2^n\}.$$

With high probability, $\mathcal{A}$ can be covered by at most $(6/\delta)^{c_2 n \delta}$ translates of the ball $\frac{C_1 \sqrt{n}}{\delta} B_2^n$.

Moreover, there exists a **non-random** subset $\mathcal{N} \subset B_2^n$ such that

$$A(B_2^n) \subset \bigcup_{y \in A(\mathcal{N})} (y + \frac{C \sqrt{n}}{\delta} B_2^n)$$

with probability at least $1 - 4 \exp(-\delta n/8)$. And $|\mathcal{N}| \leq (6/\delta)^{13n\delta}$.

Heavy tails

Light tails: with high probability,

$$\|A^{-1}\| = 1/s_{\min}(A) \sim \sqrt{n} \quad \text{and} \quad \|A\| = s_{\max}(A) \sim \sqrt{n}.$$

Heavy tails (two finite moments): with high probability,

$$\|A^{-1}\| = 1/s_{\min}(A) \sim \sqrt{n} \quad \text{and} \quad \|A\| = s_{\max}(A) \not\sim \sqrt{n}.$$

✓

???

2. Norm regularization

Questions:

1. Can we regularize the norm correcting just a small fraction of the entries of A ?
2. What in the structure of a heavy-tailed matrix causes norm to blow up from the “ideal” order $O(\sqrt{n})$?

Local regularization: $A \mapsto \bar{A}$, such that

- $\bar{A}$ differs from A in a small $\varepsilon n \times \varepsilon n$ sub-matrix
- $\|\bar{A}\| \lesssim \sqrt{n}$

Theorem (with R. Vershynin, informal statement)

Let A be a large enough random square matrix with i.i.d. elements. Local regularization is possible with high probability $\iff$
 $\mathbb{E}A_{ij} = 0$ and $\mathbb{E}A_{ij}^2$ is bounded.

Norm regularization: local obstructions

Theorem (Part 1: local obstructions)

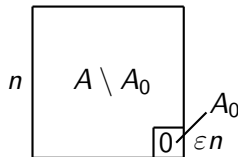
Let $A = (A_{ij})_{n \times n}$ has i.i.d. entries, such that $\mathbb{E}A_{ij} = 0$, $\mathbb{E}A_{ij}^2 = 1$. For any $\varepsilon \in (0, 0.5]$,

with probability $\geq 1 - 11e^{-\varepsilon n/12}$

there exists an $\varepsilon n \times \varepsilon n$ sub-matrix $A_0 \subset A$:

$$\|A \setminus A_0\| \leq C_\varepsilon \sqrt{n}, \quad C_\varepsilon = C \cdot \frac{\ln(\varepsilon^{-1})}{\sqrt{\varepsilon}}$$

- log-optimal dependence of on size ε
- can consider any $\varepsilon < 1$ in trade of larger constants
- inconstructive: does not identify A_0



$A \setminus A_0 = \text{zero}$ out
all entries in A_0

Norm regularization: global obstructions

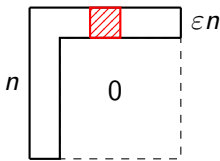
Theorem (Part 2: global obstructions)

Let $A = (A_{ij})_{n \times n}$ has i.i.d. entries, such that $\mathbb{E}A_{ij} \neq 0$ or $\mathbb{E}A_{ij}^2 = \infty$. Then for any $\delta \in (0, 1)$

$$\min \frac{\|\tilde{A}_n\|}{\sqrt{n}} \xrightarrow{a.s.} \infty \quad \text{as } n \rightarrow \infty.$$

Here the minimum is with respect to the matrices $\tilde{A}_n$ obtained by any modification of any $\delta n \times \delta n$ submatrix of A_n .

Turns out that with probability $1 - e^{-\varepsilon n}$ every small $\varepsilon n \times \varepsilon n$ submatrix has norm $\gg \sqrt{n}$.



3. Constructive regularization

Question:

1. Can we regularize the norm correcting just a small fraction of the entries of A ?

Yes, iff $\mathbb{E}A_{ij} = 0$, $\mathbb{E}A_{ij}^2 = 1$.

2. What in the structure of a heavy-tailed matrix causes norm to blow up from the “ideal” order $O(\sqrt{n})$?

Or, how to perform local regularization constructively?

Example (more than 2 moments)

Let $A = (A_{ij})_{n \times n}$ has i.i.d. entries, s.t. $\mathbb{E}A_{ij} = 0$, $\mathbb{E}|A_{ij}|^{2+\gamma} \leq 1$. For any $\gamma \in (0, 1]$ and n large enough, with high probability, zeroing out $n^{1-\gamma/9}$ largest entries of A leads to the matrix $\tilde{A}$:

$$\|\tilde{A}\| \leq 8\sqrt{n}.$$

If we have just finite *2nd* moment...

Example (failure of individual corrections approach)

Consider scaled Bernoulli matrix $A_{ij} \sim \sqrt{n} \cdot \text{Ber}(\frac{1}{n})$.

- There is a row with at least $\log n / \log \log n$ non-zero elements w.h.p. So, norm regularization is needed, as

$$\|A\| \geq \|A_i\|_2 \gg \sqrt{n}$$

- Entries are $\{0, \sqrt{n}\}$, so looking at the size only, we can only delete all or nothing.
- There are too many non-zero entries to fit in $\varepsilon n \times \varepsilon n$ sub-matrix

Need to use some information about entries locations with respect to each other (in given realization), such as **heavy rows/columns**.

3-b) Constructive regularization for Bernoulli

So, **what** can we change in a Bernoulli matrix A to get $\bar{A}$?

This is known:

- (Feige-Ofek) zero out all rows and columns of A with more than $10d$ non-zero elements,
- (Le-Levina-Vershynin) reweight or zero out some elements s.t. sum of elements in every row and column is at most $10d$,

where $d := \mathbb{E}\{\text{number of non-zero elements in a row/column}\}$.

And there are only few heavy rows and columns.

Questions:

1. Can we zero out just a small sub-matrix?
2. What is the **structure** of the bad part (to be deleted)?

Adjacency matrices of random graphs

Matrix A can be viewed as an adjacency matrix of the random graph $G(n, p_{ij})$, let $\max p_{ij} = O(\frac{1}{n})$.

So,

- (Feige-Ofek) delete all high-degree vertices in the graph (i.e. with degree $> 10 \cdot$ expected degree)
- (Le-Levina-Vershynin) reweight or delete some of the edges adjacent to high-degree vertices to make all degrees bounded

Questions:

1. To zero-out a small sub-matrix =
find a small sub-graph locating all deletions
2. Structure of the “bad” part =
how simple is the deleted sub-graph?



after_threshold.png

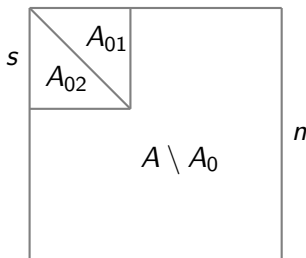
Theorem

Let A be adjacency matrix of random graph $G = G(n, p_{ij})$; $d = \max np_{ij} \geq 5$. Let us *delete a subgraph on all vertices with degrees $\geq 12d$ and their 1-neighbors*. Then, with high probability,

- resulting $\bar{A}$ has well-bounded norm

$$\|\bar{A} - \mathbb{E}A\| \lesssim \sqrt{d},$$

- deleted subgraph contains at most $s \lesssim 10ne^{-d}$ vertices
- we can direct the edges inside it without cycles s.t. every vertex will have a finite number of the incoming edges.



$A \setminus A_0$: concentration

A_{02} : rows with const non-zeros

A_{01} : col's with const non-zeros

3-c) Constructive regularization in general case

Let $A = (A_{ij})_{n \times n}$ has i.i.d. entries, such that $\mathbb{E}A_{ij} = 0$, $\mathbb{E}A_{ij}^2 = 1$.

Ideally, we would like to have something like...

Conjecture

Let A_i and A'_i denote the i -th row and column vector of A respectively. Let $\bar{A}$ be the matrix that obtained from A by zeroing out all rows and columns that have *too large L_m -norm* for some m (e.g., $m = 2$):

$$\|A_i\|_m \geq C\mathbb{E}\|A_i\|_m, \quad \|A'_i\|_m \geq C\mathbb{E}\|A'_i\|_m.$$

Then with probability $1 - o(1)$ the operator norm satisfies

$$\|\bar{A}\| \leq C'\sqrt{n}.$$

Quantiles and regularization process

Let $A = (A_{ij})_{n \times n}$ has i.i.d. entries, such that $\mathbb{E}A_{ij} = 0$, $\mathbb{E}A_{ij}^2 = 1$.

Definition (2^{-k} quantiles)

Denote as 2^{-k} -quantiles of $|A_{ij}|$ points q_k , such that

$$\mathbb{P}\{|A_{ij}| > q_k\} = 2^{-k}.$$

For every row, expected number of $A_{ij} : |A_{ij}| \in (q_{k-1}, q_k]$ is $2^{-k}n$.

Regularization procedure:

- delete too large entries
- delete rows/columns that contain $\gg 2^{-k}n$ elements between q_{k-1} and q_k for some k

Constructive regularization theorem

Theorem

Let $A = (A_{ij})_{n \times n}$ has i.i.d. symmetric entries, s.t. $\mathbb{E}A_{ij}^2 = 1$.

Let q_k be 2^{-k} -quantiles of the distribution of $|A_{ij}|$. Let $\bar{A}$ be the original matrix A after deletion of

- elements A_{ij} such that $|A_{ij}| > q_{k_1}$
- rows and columns of A that contain more than $(c_\varepsilon 2^{-k} n)$ elements such that

$$q_{k-1} < |A_{ij}| \leq q_k \text{ for some } k \leq k_1$$

Then, with probability $1 - 9n^{-9}$

- this regularization is local (in at most $4\varepsilon n$ rows and columns)
 - $\|\bar{A}\| \lesssim c_\varepsilon \sqrt{n} \sqrt{\ln \ln n}$.

For any $\varepsilon \in (0, 1/2)$ we can take $k_1 = \lceil \log_2 \frac{25n}{\varepsilon} \rceil$, $c_\varepsilon = 50/\varepsilon$.

NEXT: SOME PROOF IDEAS...

Proof ideas: invertibility

If A_{ij} **mean zero subgaussian**: $\mathbb{P}\{s_n \leq \varepsilon n^{-1/2}\} \leq L\varepsilon + e^{-un}$

Theorem (R.-Tikhomirov)

Let $A = (A_{ij})_{n \times n}$ has i.i.d. entries, s.t. $\mathbb{E}A_{ij} = 0$, $\mathbb{E}A_{ij}^2 = 1$:

$$\mathbb{P}\{s_{\min}(A) \leq \varepsilon n^{-1/2}\} \leq L_1 \varepsilon + e^{-u_1 n} \text{ for any } \varepsilon \in (0, 1)$$

$L_1, u_1 > 0$ are constants depending only on the distribution of A_{ij} .

Intermediate result - **covering numbers for random ellipdoids in $\mathbb{R}^n$** :

random ellipsoid $\mathcal{A} := \{Ax : x \in B_2^n\}$, A is as above.

With high probability, $\mathcal{A}$ can be covered by at most $(6/\delta)^{c_2 n \delta}$ translates of the ball $\frac{C_1 \sqrt{n}}{\delta} B_2^n$.

Subgaussian case, idea of the proof

$$s_n(A) := s_{\min}(A) = \min_{x \in S^{n-1}} \|Ax\|$$

Approximation by the ε -net $\mathcal{N} \subset S^{n-1}$. For any $x \in S^{n-1}$ find the closest $y \in \mathcal{N}$:

$$\begin{aligned}\|Ax\| &\geq \|Ay\| - \|A(x - y)\| \\ &\geq \|Ay\| - \|A\| \|x - y\| \\ &\geq \inf_{y \in \mathcal{N}} \|Ay\| - \sqrt{n} \cdot \varepsilon \quad (\text{used } \|A\| \sim \sqrt{n})\end{aligned}$$

Then (not easy!) ε -net with the sufficiently low cardinality was found:

$$|\mathcal{N}| \sim \left(\frac{c}{\varepsilon \sqrt{n}} \right)^n$$

Heavy-tailed case, idea of the proof

$$s_n(A) := s_{\min}(A) = \min_{x \in S^{n-1}} \|Ax\|$$

Approximation by the ε -net $\mathcal{N} \subset S^{n-1}$. For any $x \in S^{n-1}$ find the closest $y \in \mathcal{N}$:

$$\begin{aligned}\|Ax\| &\geq \|Ay\| - \|A(x - y)\| \\ &\geq \|Ay\| - \|A\| \|x - y\| \\ &\geq ??? \quad (\text{possible problem } \|A\| \gg \sqrt{n})\end{aligned}$$

Challenge: obtain an estimate $\|A(x - y)\| \leq \sqrt{n}\varepsilon$, where x, y are in the same ε -net element.

So, for any x, y taken from one net element we would like to have

$$\|A(x - y)\| \leq \sqrt{n}\varepsilon$$

dilation2.pdf

Covering theorem

Theorem

For any $\delta < 1/4$ and $n > n_0(\delta)$ there exists a non-random subset $\mathcal{N} \subset B_2^n$ of cardinality at most $(6/\delta)^{13n\delta}$, such that for any $n \times n$ random matrix A with i.i.d. elements such that $\mathbb{E}A_{ij} = 0$ and $\mathbb{E}A_{ij}^2 = 1$, we have

$$A(B_2^n) \subset \bigcup_{y \in A(\mathcal{N})} \left(y + \frac{C\sqrt{n}}{\delta} B_2^n\right)$$

with probability at least $1 - 4 \exp(-\delta n/8)$.

NEXT: LOCAL REGULARIZATION OF THE NORM
(proof ideas)

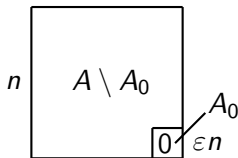
Proof ideas: norm regularization

Theorem (Local obstructions)

Let A be a $n \times n$ matrix with i.i.d. entries, $\mathbb{E}A_{ij} = 0$, $\mathbb{E}A_{ij}^2 = 1$.
Then for any $\varepsilon \in (0, \frac{1}{2}]$ with probability at least $1 - 11 \exp(-\varepsilon n/12)$ there exists an $\varepsilon n \times \varepsilon n$ sub-matrix $A_0 \subset A$:

$$\|A \setminus A_0\| \leq C_\varepsilon \sqrt{n}, \quad C_\varepsilon = C \cdot \frac{\ln(\varepsilon^{-1})}{\sqrt{\varepsilon}}$$

Here, $A \setminus A_0$ is a matrix we obtain by zeroing out all elements of A , that belong to sub-matrix A_0 :



Dependence on ε

Optimal dependence would be $C_\varepsilon = O(\frac{1}{\sqrt{\varepsilon}})$.

To see this, consider

- $\varepsilon \lesssim \frac{1}{n}$ and $\|A\| = O(n) = O(\sqrt{n}/\sqrt{\varepsilon})$ with probability $1/2$, or
- any $\varepsilon \leq 1/2$ and Bernoulli matrix A with rare $\frac{\sqrt{n}}{\sqrt{\varepsilon}}$ spikes

Our argument gives **log-optimal dependence** $C_\varepsilon = O(\frac{\ln(\varepsilon^{-1})}{\sqrt{\varepsilon}})$

Example (Bernoulli case)

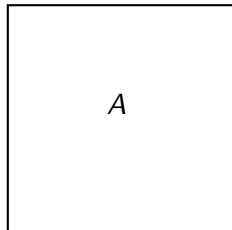
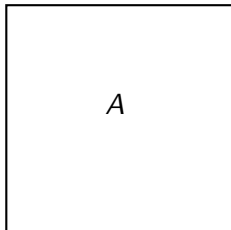
However, in "good Bernoulli case" dependence is better. Let A be a square matrix with i.i.d. elements distributed like

$$A_{ij} = \begin{cases} \frac{1}{\sqrt{p}} & \text{with probability } p \\ 0 & \text{otherwise} \end{cases}, \quad p \cdot n = d = O(1) \geq 4$$

Then the theorem holds with $C_\varepsilon = O(\ln(\varepsilon^{-1}))$.

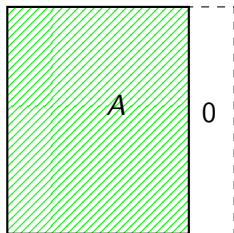
Observation 0: εn columns cut

It is enough to show that εn -columns cut regularizes the norm:

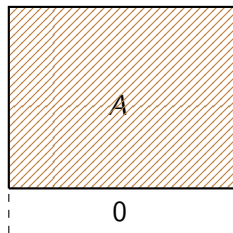


Observation 0: εn columns cut

It is enough to show that εn -columns cut regularizes the norm:



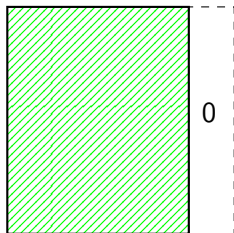
$$\|\text{green}\| \leq \sqrt{n}$$



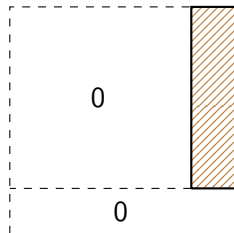
$$\|\text{brown}\| \leq \sqrt{n}$$

Observation 0: εn columns cut

It is enough to show that εn -columns cut regularizes the norm:



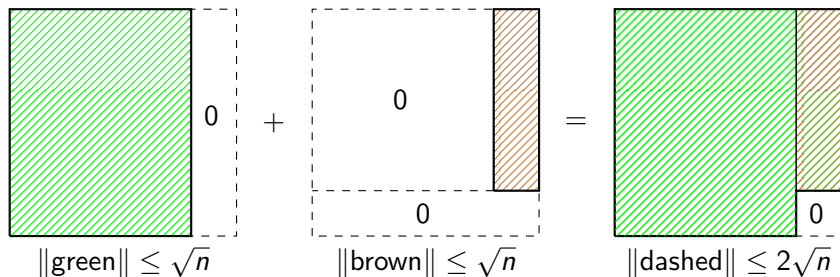
$$\|\text{green}\| \leq \sqrt{n}$$



$$\|\text{brown}\| \leq \sqrt{n}$$

Observation 0: εn columns cut

It is enough to show that εn - columns cut regularizes the norm:


$$\begin{array}{ccc} \begin{array}{|c|} \hline \text{green} \\ \hline \end{array} & + & \begin{array}{|c|} \hline \text{brown} \\ \hline \end{array} & = & \begin{array}{|c|} \hline \text{dashed} \\ \hline \end{array} \\ \|\text{green}\| \leq \sqrt{n} & & \|\text{brown}\| \leq \sqrt{n} & & \|\text{dashed}\| \leq 2\sqrt{n} \end{array}$$

Three norms

Definition

- Operator norm

$$\|A\| = \|A : l_2 \rightarrow l_2\| = \sup_{\|x\|_2=1} \|Ax\|_2$$

- Infinity to 2 (cut norm)

$$\|A\|_{\infty \rightarrow 2} = \|A : l_\infty \rightarrow l_2\| = \max_{x \in \{-1,1\}^n} \|Ax\|_2$$

- 2 to infinity (maximum row norm)

$$\|A\|_{2 \rightarrow \infty} = \|A : l_2 \rightarrow l_\infty\| = \max_i \|A_i\|_2,$$

where A_i , $i = 1, \dots, n$ denote rows of matrix A .

Example

For gaussian matrix (i.i.d. $N(0,1)$ entries) we have:

$$\|A\|_{2 \rightarrow \infty} \sim \sqrt{n}, \quad \|A\|_{\infty \rightarrow 2} \sim n, \quad \|A\| \sim \sqrt{n}$$

"Ideal" norm relation?

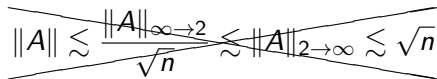
$$\|A\| \lesssim \frac{\|A\|_{\infty \rightarrow 2}}{\sqrt{n}} \lesssim \|A\|_{2 \rightarrow \infty} \lesssim \sqrt{n}$$

Example

For gaussian matrix (i.i.d. $N(0,1)$ entries) we have:

$$\|A\|_{2 \rightarrow \infty} \sim \sqrt{n}, \quad \|A\|_{\infty \rightarrow 2} \sim n, \quad \|A\| \sim \sqrt{n}$$

"Ideal" norm relation?


$$\|A\| \lesssim \frac{\|A\|_{\infty \rightarrow 2}}{\sqrt{n}} \lesssim \|A\|_{2 \rightarrow \infty} \lesssim \sqrt{n}$$

Not true :) Instead,

$$\|A_{J_3^c}\| \lesssim \frac{\|A_{J_2^c}\|_{\infty \rightarrow 2}}{\sqrt{n}} \lesssim \|A_{J_1^c}\|_{2 \rightarrow \infty} \lesssim \sqrt{n},$$

where J_1, J_2, J_3 are small subsets of columns that we zero out
($J_1 \subset J_2 \subset J_3$ with cardinalities $|J_i| \leq \varepsilon n$)

The $2 \rightarrow \infty$ norm: damping

Lemma

Consider an $n \times n$ random matrix A with i.i.d. entries A_{ij} which have mean zero, unit variance and $|A_{ij}| \leq \frac{\sqrt{n}}{2}$ a.s. Let $\varepsilon \in (0, 1/2]$. Then with probability at least $1 - e^{-\varepsilon n}$, there exists a subset $J_1 \in [n]$ with cardinality $|J_1| \leq \varepsilon n$ such that

$$\|A_{J_1^c}\|_{2 \rightarrow \infty} \leq C \sqrt{\ln \varepsilon^{-1}} \cdot \sqrt{n}.$$

Warning: we cannot just cut columns with large elements!

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	*	*	

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*			

Damping: Bernoulli example

Idea: we construct a **diagonal** matrix of weights that regularizes each row

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & & & & \\ & \delta_1 & & & \\ & & 0 & & \\ & & & 0 & \\ & & & & \delta_1 \end{bmatrix} = \begin{bmatrix} 0 & \delta_1 & 0 & 0 & \delta_1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1-st row: damping with the weight $0 < \delta_1 < 1$

Damping: Bernoulli example

Idea: we construct a **diagonal** matrix of weights that regularizes each row

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ \color{red}{0} & \color{red}{0} & \color{red}{0} & \color{red}{0} & \color{red}{0} \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & & & & \\ & \delta_1 & & & \\ & & 0 & & \\ & & & 0 & \\ & & & & \delta_1 \end{bmatrix} = \begin{bmatrix} 0 & \delta_1 & 0 & 0 & \delta_1 \\ \color{red}{0} & \color{red}{0} & \color{red}{0} & \color{red}{0} & \color{red}{0} \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2-nd row: all good

Damping: Bernoulli example

Idea: we construct a **diagonal** matrix of weights that regularizes each row

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & & & & \\ & \delta_1^2 & & & \\ & & \delta_1 & & \\ & & & 0 & \\ & & & & \delta_1 \end{bmatrix} = \begin{bmatrix} 0 & \delta_1 & 0 & 0 & \delta_1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \delta_1 & \delta_1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

3-rd row: damping with the weight $0 < \delta_1 < 1$

Damping: Bernoulli example

Idea: we construct a **diagonal** matrix of weights that regularizes each row

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ \color{red}{1} & \color{red}{1} & \color{red}{0} & \color{red}{0} & \color{red}{1} \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \color{red}{\delta_2} & & & & \\ & \color{red}{\delta_1^2 \delta_2} & & & \\ & & \delta_1 & & \\ & & & 0 & \\ & & & & \color{red}{\delta_1 \delta_2} \end{bmatrix} = \begin{bmatrix} 0 & \delta_1 & 0 & 0 & \delta_1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \delta_1 & \delta_1 & 0 & 0 \\ \color{red}{\delta_2} & \color{red}{\delta_2} & \color{red}{0} & \color{red}{0} & \color{red}{\delta_2} \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

4-th row: damping with the weight $0 < \delta_2 < \delta_1 < 1$

Damping: Bernoulli example

Idea: we construct a **diagonal** matrix of weights that regularizes each row

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ \color{red}{1} & \color{red}{0} & \color{red}{0} & \color{red}{0} & \color{red}{0} \end{bmatrix} \begin{bmatrix} \delta_2 & & & & \\ & \delta_1^2 \delta_2 & & & \\ & & \delta_1 & & \\ & & & 0 & \\ & & & & \delta_1 \delta_2 \end{bmatrix} = \begin{bmatrix} 0 & \delta_1 & 0 & 0 & \delta_1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \delta_1 & \delta_1 & 0 & 0 \\ \delta_2 & \delta_2 & 0 & 0 & \delta_2 \\ \color{red}{1} & \color{red}{0} & \color{red}{0} & \color{red}{0} & \color{red}{0} \end{bmatrix}$$

5-th row: all good

Damping: Bernoulli example

Idea: we construct a **diagonal** matrix of weights that regularizes each row

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_2 & & & & \\ & \delta_1^2 \delta_2 & & & \\ & & \delta_1 & & \\ & & & 0 & \\ & & & & \delta_1 \delta_2 \end{bmatrix} = \begin{bmatrix} 0 & \delta_1 & 0 & 0 & \delta_1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \delta_1 & \delta_1 & 0 & 0 \\ \delta_2 & \delta_2 & 0 & 0 & \delta_2 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2-nd column has small weight: to be deleted

Damping: Bernoulli example

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_2 & & & & \\ & \delta_1^2 \delta_2 & & & \\ & & \delta_1 & & \\ & & & 0 & \\ & & & & \delta_1 \delta_2 \end{bmatrix} = \begin{bmatrix} 0 & \delta_1 & 0 & 0 & \delta_1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \delta_1 & \delta_1 & 0 & 0 \\ \delta_2 & \delta_2 & 0 & 0 & \delta_2 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Proposition

Let $\varepsilon \in (0, 1]$ and A is our matrix. Then with high probability there exists a diagonal weight matrix $D = (d_{ii})_{i=1}^n$, $d_i \in (0, 1)$, such that

- (1) $\|AD\|_{2 \rightarrow \infty} \leq C\sqrt{\ln \varepsilon^{-1}}\sqrt{n}$
- (2) $\mathbb{E}(d_{11} \cdot d_{22} \cdot \dots \cdot d_{nn}) \geq \exp(-\varepsilon n)$

- Condition (2) implies that there all but εn columns have weights d_{ii} 's such that: $d_{ii} > e^{-2}$. We can cut the rest!

Damping: general distribution case

Lemma (Discretization)

Consider a non-negative, continuous random variable X . There exists a non-negative random variable X' satisfying the following.

1. $\mathbb{E}X' \leq 4\mathbb{E}X$.
2. $X' \geq X$ almost surely
3. X' is a sum of scaled, *independent Bernoulli* random variables:

$$X' = \sum_{k=0}^{\infty} q_k \xi_k$$

where q_k are non-negative numbers and ξ_k are independent $\text{Ber}(2^{-k})$ random variables.

Step 2: $\|\cdot\|_{\infty \rightarrow 2}$ norm - playing with the signs

Reminder: we are proving

$$\|A_{J_3^c}\| \lesssim \frac{\|A_{J_2^c}\|_{\infty \rightarrow 2}}{\sqrt{n}} \lesssim \|A_{J_1^c}\|_{2 \rightarrow \infty} \lesssim \sqrt{n},$$

Lemma

Let A be an $n \times n$ random matrix whose entries are independent, *symmetric* random variables. Then

$$\|A\|_{\infty \rightarrow 2} \leq C\sqrt{n}\|A\|_{2 \rightarrow \infty}$$

with probability at least $1 - e^{-n}$.

Rough idea: condition on $|A_{ij}|$, and consider linear combination of Rademacher random variables ($\gamma := \pm 1$ with probability $1/2$)

Corollary

If $J_1 \subset [n]$ be a random subset, which is independent of the signs of the entries of A , then with the same high probability

$$\|A_{J_1^c}\|_{\infty \rightarrow 2} \leq C\sqrt{n}\|A_{J_1^c}\|_{2 \rightarrow \infty}$$

So, $J_1 = J_2$, there are no loss on Step 2.

Removing symmetry assumption:

- Note that for Lemma basic anti-symmetrization inequality will do (norm is a **convex** function)

$$\mathbb{E}\varphi(\|\sum_i X_i\|) \leq \mathbb{E}\varphi(2\|\sum_i \gamma_i X_i\|) \quad (\text{from Ledoux-Talagrand})$$

- However, for Corollary (columns deletion makes it **non-convex**) more delicate argument is needed.

Step 3: $\|\cdot\|$ norm - Grothendieck-Pietsch factorization

Standard estimate: $\frac{1}{\sqrt{n}}\|B\|_{\infty \rightarrow 2} \leq \|B\| \leq \|B\|_{\infty \rightarrow 2}$

We want: $\|A_{J_3^c}\| \lesssim \frac{1}{\sqrt{n}}\|A_{J_2^c}\|_{\infty \rightarrow 2}$ with high probability

Theorem (Grothendieck-Pietsch, sub-matrix version)

Let B be a $n \times n_1$ real matrix and $\delta > 0$. Then there exists $J \subset [n_1]$ with $|J| \geq (1 - \varepsilon)n_1$ such that

$$\|B_{[n] \times J}\| \leq \frac{2\|B\|_{\infty \rightarrow 2}}{\sqrt{\varepsilon n_1}}.$$

We use it with $n_1 = (1 - \varepsilon)n$ to find $|J| \geq (1 - 2\varepsilon)n$, such that

$$\|A_{[n] \times J}\| \leq \frac{2\|A \setminus A'\|_{\infty \rightarrow 2}}{\sqrt{\varepsilon n}} \leq \frac{C_\varepsilon n}{\sqrt{\varepsilon n}} \leq \frac{C_\varepsilon}{\sqrt{\varepsilon}} \sqrt{n}.$$

Fighting for a good C_ε

Solution (for bounded entries): consider only "small" entries of the matrix $|A_{ij}| \lesssim \sqrt{n}$, then on Step 1 $\|A \setminus A'\|_{2 \rightarrow \infty} \leq C \sqrt{\ln(\varepsilon^{-1})n}$.

Hence, for a matrix A such that $\mathbb{E}A_{ij} = 0$, $\mathbb{E}A_{ij}^2 \leq 1$, $|A_{ij}| \leq \frac{\sqrt{n}}{2}$ a.s.:

$$\|A \setminus A'\| \leq C \frac{\sqrt{\ln(\varepsilon^{-1})}}{\sqrt{\varepsilon}} \sqrt{n}.$$

General case:

$$A = A \cdot \mathbb{1}_{\{|A_{ij}| \lesssim \sqrt{n}\}} + A \cdot \mathbb{1}_{\{\sqrt{n} \lesssim |A_{ij}| \lesssim \frac{\sqrt{n}}{\sqrt{\varepsilon}}\}} + A \cdot \mathbb{1}_{\{\frac{\sqrt{n}}{\sqrt{\varepsilon}} \lesssim |A_{ij}|\}}$$

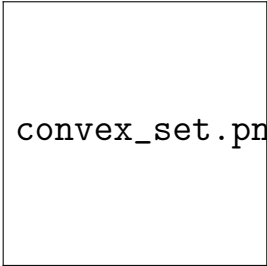


sparsity and size
(most non-zero elements
belong to sparse rows)



very sparse
(εn non-zero
elements)

THANKS FOR YOUR ATTENTION!



convex_set.png

Applications to community detection

A - adjacency matrix of Erdős-Rényi random graph $G(n, p_{ij})$.

Stochastic block model:

$$\mathbb{E}A = \left[\begin{array}{cc|cc} p & p & s & s \\ p & p & s & s \\ \hline s & s & p & p \\ s & s & p & p \end{array} \right]$$

2comm.pdf

Spectral method for community detection is based on the idea:

1. $\mathbb{E}A$ eigenstructure reveals communities,
2. $\text{eigenstructure}[A] \sim \text{eigenstructure}[\mathbb{E}A]$.

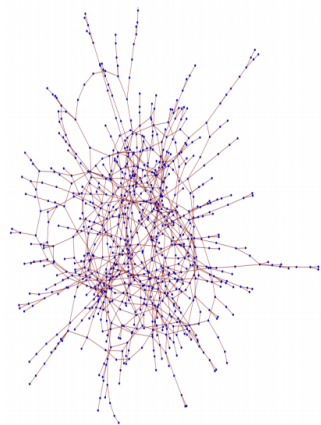
So, eigenstructure of A (observed) reveals communities too.

Condition 2 is satisfied only for **dense** graphs. Idea for **sparse** graphs: **regularize graph locally** to impose $\|A - \mathbb{E}A\| \sim 0$.

Obstructions for random graphs

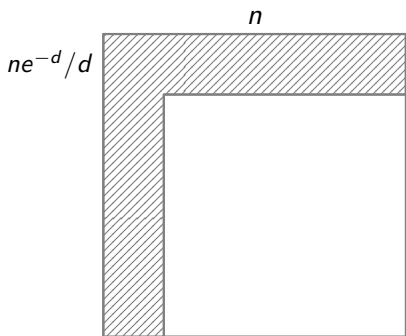
Sparse graphs: maximal expected degree $d := \max p_{ij} n \ll \log n$.

[Feige-Ofek] **obstructions** to $\|A - \mathbb{E}A\| \sim 0$ are **few high degree vertices** of the graph.

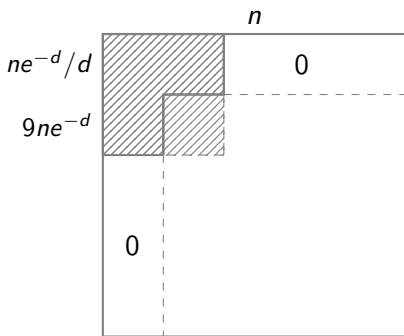


For the regularization it is enough to

- Feige-Ofek: delete all high-degree vertices (degree $> 10d$)
- Le-Levina-Vershynin: reweight or delete some of the edges adjacent to high-degree vertices (to make all the degrees $< 10d$)

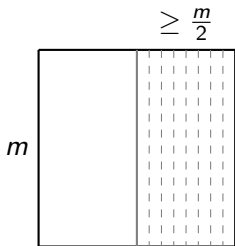


Step 1: heavy rows/columns are dashed

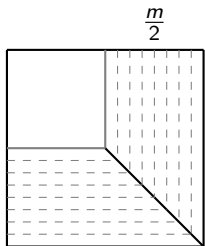


Step 2: A_0 is dashed

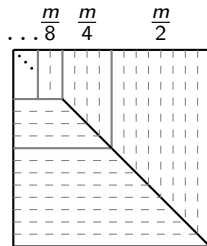
Construction of A_0



Step 1



Step 1 + symmetry



Inductive steps

Inductive decomposition A_0 . Dotted lines – “light” rows or columns

Theorem (Part 2: global obstructions)

Let A is an $n \times n$ matrix with i.i.d. entries, such that

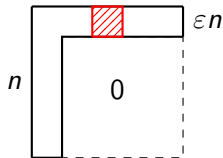
- $\mathbb{E}A_{ij}^2 \geq M$,
- $|A_{ij}| \leq \sqrt{n}$ almost surely.

If $M = M(C, \varepsilon)$ is a large enough constant, then **any** $\varepsilon n \times \varepsilon n$ sub-matrix A_0 has large norm

$$\|A_0\| \geq C\sqrt{n},$$

with probability at least $1 - \exp(-\varepsilon n)$.

So, if we were to cut some part for regularization, we need to cut almost everything! No $\varepsilon n \times \varepsilon n$ sub-matrix can survive.



Proof idea

Frobenius norm $\|A_0\|_F^2 := \sum_{i=1}^n s_i^2 \leq n \cdot \max s_i^2 = n \cdot \|A_0\|^2$

- it's enough to show that Frobenius norm is large

$$\|A_0\|_F \geq Cn - ?$$

- split elements onto levels "by size":

$$\|A_0\|_F^2 = \sum_{A_{ij} \in A_0} A_{ij}^2 = \sum_{k=0}^{\infty} \sum_{A_{ij} \in A_0} a_{ij}^2 \mathbb{1}_{\{2^k \leq a_{ij}^2 < 2^{k+1}\}}$$

- argue that the majority of the levels in any $\varepsilon n \times \varepsilon n$ sub-block contain many non-zero elements (use Chernoff's inequality).

Done!

Damping for each row

So, enough to show that for every row A_i exists $D^i = (d_{11}^i, \dots, d_{nn}^i)$:

- $\sum_j d_{jj}^i \cdot A_{ij}^2 \leq C_\varepsilon n$
- $\mathbb{E}(d_{11}^i \cdot d_{22}^i \cdot \dots \cdot d_{nn}^i) \geq e^{-\varepsilon}$

For Bernoulli matrix:

$$\begin{cases} d_{jj}^i := 0, & \text{if } A_{ij} = 0 \\ d_{jj}^i := 1, & \text{if } \|A_i^2\|_1 \leq Cn \\ d_{jj}^i := \frac{Cn}{\|A_i^2\|_1}, & \text{otherwise} \end{cases}$$

where $A_i^2 := (A_{i1}^2, \dots, A_{in}^2)$.

For general case:

Naive regularization ($d_{jj}^i := \frac{\text{expected norm}}{\text{real norm}}$) would not work

Proof of Step 2

Lemma

Let A be an $n \times n$ random matrix whose entries are independent, *symmetric* random variables. Then

$$\|A\|_{\infty \rightarrow 2} \leq C\sqrt{n}\|A\|_{2 \rightarrow \infty}$$

with probability at least $1 - e^{-n}$.

We want to show:

$$\max_{\{ -1, 1 \}^n} \|Ax\|_2^2 \leq Cn \max_{\text{rows}} \|A_i\|_2^2 \quad \text{w/high probability}$$

Enough to show: for each $x \in \max_{\{ -1, 1 \}^n}$

$$\|Ax\|_2^2 \leq Cn \max_{\text{rows}} \|A_i\|_2^2 \quad + \text{ union bound}$$

$$\|Ax\|_2^2 \leq Cn \max \|A_i\|_2^2 - ?$$

Left hand side $\|Ax\|_2^2 = \sum \xi_i^2$, where

$$\xi_i = \langle A_i, x \rangle = \sum_j A_{ij}x_j = \sum_j A_{ij}\gamma_{ij}x_j = \sum_j A_{ij}\gamma_{ij}.$$

Linear combination of ± 1 symmetric random variables γ_{ij} - they are subgaussian. Bernstein for subgaussians: ξ_i is also subgaussian with $\|\xi_i\|_{\psi_2}^2 = \sum_j A_{ij}^2 = \|A_i\|_2^2$.

ξ_i - subgaussian $\therefore \xi_i^2$ - subexponential

Concentration for sum of subexponentials:

$$\|Ax\|_2^2 = \sum \xi_i^2 \leq C \cdot n \|\xi_i\|_{\psi_2}^2 \leq Cn \|A_i\|_2^2.$$

Done!